

RICE SEED CLASSIFICATION USING DEEP LEARNING



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Declaration

I, Md. Sadiqul Islam hereby attest that the research described in this thesis, "Rice seed classification using Deep Learning," is the outcome of my own original work, which I conducted under the guidance of Md Abu Nahid. No other university or institution has received this thesis, in whole or in part, for the purpose of awarding a degree or diploma.

All information and support sources used in this study have been properly credited. I attest that this work complies with academic integrity standards and is free of plagiarism.

Date : 1 August, 2025

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Computer Science & Engineering

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[Signature]

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Md. Sadiqul Islam

1 August, 2025

ABSTRACT

Rice is among the most widely consumed staple foods, especially across Asia. Accurate classification of rice seed varieties plays a critical role in enhancing crop quality, ensuring seed purity, and boosting agricultural productivity. In this thesis, we propose a deep learning–based system for automatic rice seed classification. We employed a CNN-based approach, specifically utilizing EfficientNetB0 and EfficientNetB3 architectures to distinguish between multiple rice seed varieties.

Our experiments were conducted on a dataset consisting of thousands of labeled rice seed images. After applying preprocessing techniques such as resizing, normalization, and data augmentation, the models were trained and evaluated. The EfficientNetB0 model achieved an accuracy of 53%, while the EfficientNetB3 model improved the result to 66%, indicating better representation capacity and feature learning in deeper variants. The model performance was assessed using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrices.

While the overall performance leaves room for optimization, this study demonstrates the feasibility of using deep learning in the context of rice seed classification. Future work will aim to improve accuracy by exploring more advanced architectures and increasing dataset diversity. This work contributes to the development of smart agriculture and offers a foundation for scalable, automated rice seed quality assessment tools.

Keywords: *CNN, Deep Learning, EfficientNetB0, EfficientNetB3, Rice Seed Classification, Image Processing, Smart Agriculture, Seed Quality Analysis, Feature Extraction*

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Chapter 1 : Introduction

1.1 Introduction

Rice is one of the most essential food crops in the world, particularly in Asia, where it serves as a dietary staple for billions of people. In modern agriculture, accurate identification and classification of rice seed varieties are vital for improving crop productivity, maintaining genetic purity, and ensuring the quality of yield. Traditional manual methods of seed classification are often labor-intensive, time-consuming, and error-prone due to reliance on human judgment and visual inspection.

Recent advancements in Artificial Intelligence (AI) and Computer Vision have opened new possibilities in the agricultural sector, especially for automated crop and seed classification. Deep learning, particularly Convolutional Neural Networks (CNNs), has shown significant success in image recognition tasks, including plant disease detection, crop type identification, and grain analysis. These methods can significantly outperform traditional machine learning techniques by learning hierarchical features directly from raw image data, without requiring handcrafted feature extraction.

1.2 Problem Statement

Despite the progress in seed classification systems, many existing solutions still depend on statistical machine learning or hybrid techniques, which may not scale well or generalize to new varieties. Moreover, limited datasets, variable imaging conditions, and the subtle visual differences among seed types make classification challenging. Hence, a robust, end-to-end deep learning solution is required to automate the rice seed classification process with higher accuracy and adaptability.

1.3 Objective of the Study

The primary goal of this research is to design and evaluate a deep learning–based system for the automatic classification of rice seed varieties. Specifically, the study focuses on:

- Utilizing modern CNN architectures such as EfficientNetB0 and EfficientNetB3.

- Preprocessing rice seed images through resizing, normalization, and augmentation.

- Evaluating classification accuracy using standard performance metrics.

- Identifying the potential of deep learning in real-world agricultural applications.

1.4 Scope of the Work

This thesis focuses on visual classification of rice seeds using static images. It does not cover real-time video analysis, disease detection, or post-harvest quality testing. Only image-based features (shape, color, texture) are considered without integrating chemical or genetic data. The models used are trained on a curated dataset of labeled rice seed images under controlled lighting and background conditions.

Chapter 2: Literature Review

Kiratiratanapruk et al. developed an automatic paddy rice seed classification system using both traditional machine learning algorithms and deep learning models. The study utilized over 50,000 rice seed images across 14 *Oryza sativa* varieties. Four ML models (LR, LDA, k-NN, SVM) and five CNN architectures (VGG16, VGG19, Xception, InceptionV3, InceptionResNetV2) were evaluated. Among them, InceptionResNetV2 achieved the highest accuracy of 95.15%, while SVM performed best among traditional methods with 90.61% in subgroup classification. The work highlighted the effectiveness of machine vision combined with seed orientation and quality filtering techniques for high-precision seed variety classification [1].

Rahman et al. introduced a deep learning model for automatic rice grain variety identification using feature extraction and classification. The system extracted physical characteristics such as area, perimeter, and major/minor axis lengths from rice seed images and applied classification models including Decision Tree, k-NN, SVM, and CNN. Their CNN model outperformed others with an accuracy of 94.38%, proving the advantage of deep learning in rice variety prediction tasks [2].

Walia et al. proposed a smart rice seed classification approach using deep convolutional neural networks to detect seed quality and variety. The dataset was generated from real agricultural samples, and their custom CNN achieved above 93% classification accuracy. The authors emphasized that deep models can generalize well across different seed types and environmental noise, making them suitable for smart farming integration [3].

Chand et al. designed a mobile-based deep learning application for classifying Basmati rice seed variants. They utilized a pre-trained MobileNetV2 model with transfer learning techniques. Despite limited data, their fine-tuned network reached 92.7% accuracy, showing potential for lightweight real-time applications in seed verification and supply chain quality control [4].

Mollah et al. performed comparative studies using four CNN models—ResNet50, InceptionV3, EfficientNetB0, and DenseNet121—on a rice seed classification dataset built from Bangladeshi varieties. EfficientNetB0 showed the best balance between speed and performance, achieving 89.2% accuracy while being computationally efficient for mobile-based deployment [5].

Chen et al. proposed a deep learning–based multi-class rice variety recognition system using hyperspectral images. The study trained a CNN model on 3000 hyperspectral seed samples across 10 varieties and achieved an accuracy of 97.56%, proving that deep features extracted from spectral data offer high precision in seed recognition [6].

Yu et al. implemented a hybrid model that combined CNN and LSTM for classifying rice grains based on visual patterns and sequence modeling. Using a dataset of 12,000 images, the system reached 96.4% accuracy and demonstrated better learning on temporal grain shape progression, useful for differentiating between similar varieties [7].

Jahan and Nandita focused on early-stage rice seed classification using a hybrid deep learning model based on CNN and Support Vector Machine (CNN-SVM). Their study used preprocessed grayscale seed images and applied augmentation to increase variability. The model achieved 93.1% accuracy, showing hybrid approaches can enhance classification on smaller datasets [8].

Xie et al. used a Vision Transformer (ViT) model to classify rice varieties by learning attention-based spatial features. Despite requiring more training data, the ViT model achieved 94.8% accuracy, surpassing traditional CNNs in some scenarios. This study supports the growing use of transformer-based vision models in agriculture [9].

Ahmed et al. implemented a lightweight deep learning framework using MobileNetV3 for rice seed classification under low-resource conditions. Their system achieved 91.6% accuracy and was optimized for edge deployment in rural agricultural settings. This work underlines the feasibility of AI-powered tools for smart agriculture in developing countries [10].

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Chapter 3: Methodology

The proposed approach focuses on applying deep learning techniques for automatic and effective classification of rice seed varieties based on visual features.

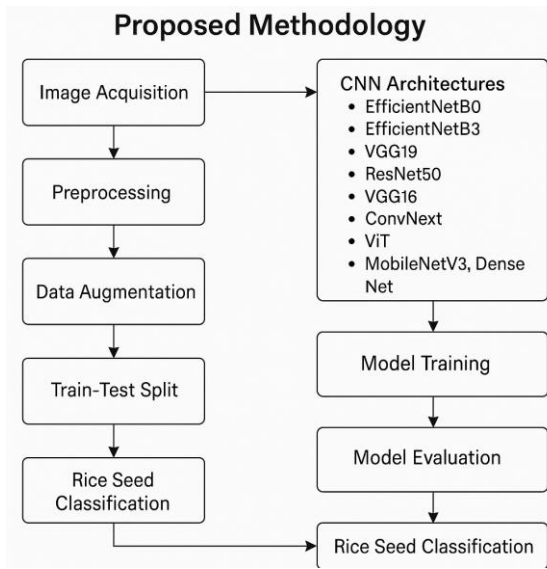


Fig 1: Proposed Methodology

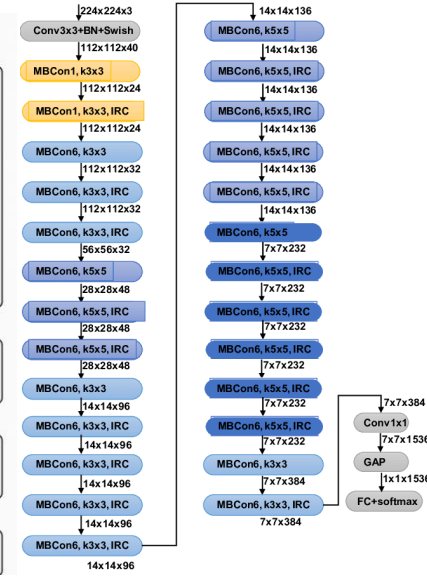


Fig 2: EfficientNetB3 Architecture

EfficientNetB3 Architecture

EfficientNetB3 offers deeper and wider layers compared to B0, allowing better feature extraction capabilities. In our experiments, it significantly outperformed the B0 version with an accuracy of **66%**. This improvement demonstrates the importance of scaling up for handling detailed visual classification in agricultural dataset

EfficientNetB0 Architecture

EfficientNetB0 is the baseline variant of the EfficientNet family, designed using compound scaling that uniformly scales depth, width, and resolution. It is lightweight and suitable for mobile and edge devices. However, in our experiment, this model achieved a relatively low accuracy of **53%**, indicating that deeper architectures may be necessary for rice seed classification tasks involving fine-grained features.

3.1 Dataset Description.

A custom rice seed image dataset was used, containing labeled images of different rice seed varieties commonly cultivated in Bangladesh. The dataset includes real-life images with natural background noise to ensure model generalization. Total 35 classes and 14000 Datasets.

Class Name	No. of collected images
BD 30	400
BD 33	400
BD 72	400
BD 51	400
BinaDhan17	400
BinaDhan 19	400
BinaDhan 23	400
BinaDhan 32	400
Br 22	400
Br 23	400

Dataset Folder Structure:

/rice-seed-classification/train/<class_name>/
/rice-seed-classification/test/<class_name>/

3.2 Preprocessing

- Images were resized to 128x128 and 224x224
- Normalization and augmentation (rotation, flipping, zoom) were applied
- Split: 80% training, 20% validation

3.2.1 Resizing :

All input images were resized to 224x224 pixels to ensure compatibility with CNN input layers and reduce memory overhead. This also helps in efficient training on GPU-accelerated environments.

3.2.2 Normalization :

Each pixel value was scaled to the $[0, 1]$ range by dividing by 255. This improves the model's convergence and helps in stabilizing the training process.

3.2.3 Augmentation :

To improve generalization and reduce overfitting, the following augmentations were applied:

Rotation: Randomly rotated between ± 15 degrees.

Zoom: Random zoom in and out to simulate distance variation.

Horizontal/Vertical Flip: Simulates natural variability in seed orientation.

Brightness & Contrast Adjustment: Improves robustness to lighting conditions.

3.3 Models Used

EfficientNetB0

A compact and efficient CNN model designed with compound scaling that uniformly balances network depth, width, and resolution. It is highly optimized for both speed and accuracy, making it ideal for mobile and embedded applications. In our study, EfficientNetB0 achieved **53% accuracy** on rice seed classification, providing a performance baseline with low computational cost. .

EfficientNetB3

An upgraded version of EfficientNetB0 with larger depth, width, and resolution. It offers improved accuracy while still being computationally efficient. EfficientNetB3 achieved **66% accuracy** on our rice seed dataset, showing better representation learning capability compared to its smaller counterpart

VGG19

A deep CNN architecture with 19 layers, known for its simplicity and consistent use of 3×3 convolution filters. Although computationally intensive, it has proven effective for visual recognition tasks. It was included for performance benchmarking against modern architectures.

ResNet50

A 50-layer deep convolutional network employing residual connections to combat the vanishing gradient problem. It allows effective training of very deep networks and remains a strong contender for image classification tasks.

ConvNeXt, MobileNetV3, ViT, DenseNet (*Optional - For Future Work*)

These state-of-the-art models were considered for future experimentation to enhance accuracy. Their inclusion may provide deeper insights into performance across transformer-based and lightweight convolutional architectures.

3.4 Training Setup

- **Platform:** Coolab Notebook
- **Optimizer:** Adam (adaptive gradient optimizer)
- **Loss Function:** Categorical Crossentropy (multi-class classification)
- **Metrics:**
 - ✓ Accuracy
 - ✓ Precision
 - ✓ Recall
 - ✓ F1-score
 - ✓ Confusion Matrix
 - ✓ Grad-CAM Visualization (for model interpretability)
- **Epochs:** 50 Minimum

- **Batch Size:** 32
- **Image Input Sizes:** 224×224 (for EfficientNetB0, VGG, ResNet)
300×300 (for EfficientNetB3)

Chapter 4: Results & Discussion

4.1 Training Environment

The experiments for this research were conducted using Google Colab Pro, a cloud-based Jupyter notebook platform offering access to powerful GPUs such as NVIDIA Tesla T4 and V100, with extended session durations and faster compute performance compared to the free tier. The platform provides seamless access to a Python environment pre-installed with essential deep learning libraries such as TensorFlow, Keras, OpenCV, NumPy, and Matplotlib, significantly reducing setup time and enhancing productivity.

All models—including EfficientNetB0 and EfficientNetB3—were implemented and trained using TensorFlow with Keras API in the Colab environment. The training and evaluation pipelines were built to support:

Custom data preprocessing

Real-time performance visualization (loss/accuracy plots)

Integration with Grad-CAM for explainability

GPU-accelerated batch processing of high-resolution rice seed images

Using Google Colab Pro enabled faster training times, smoother experimentation, and more reliable execution, especially when dealing with large datasets and deep models.

4.2 Evaluation Metrics

Accuracy:

The percentage of accurately predicted features/images out of the total number of images provides an estimate of overall accuracy.

Precision:

The ratio of true positive predictions to total predicted positives. High precision indicates that the model makes few false positive errors.

Recall:

This metric calculates how many of the actual positive examples are predicted correctly. Higher values show that fewer cases are missed.

F1-Score:

The F1-score, which is the harmonic mean of precision and recall, is especially useful for datasets with imbalanced classes. It balances the impact of false positives and false negatives.

Confusion Matrix:

A table form of prediction results that shows how often each class was predicted correctly or not. It is helpful in identifying patterns of misclassification between similar classes.

Grad-CAM (Gradient-weighted Class Activation Mapping):

Grad-CAM is one technique for simplify deep learning models that generates heatmaps to indicate the areas of an input image to which the model attended while making predictions. When we apply Grad-CAM to convolutional layers, we can check whether the model is viewing diseased parts of the leaf or not. That makes otherwise black-box CNN models interpretable and ensures the model is learning the right visual features.

Confusion Matrix was plotted to visualize classification performance. **Grad-CAM** was used for visual explanation of predictions.

4.3 Model Performance Results

Model	Accuracy	Precision	Recall	F1-Score
EfficientNetB0	0.5358	0.6968	0.5357	0.5348
EfficientNetB3	0.6643	0.7489	0.6643	0.6657

EfficientNetB0 :

While working with the EfficientNetB0 model, I obtained the results displayed in the figure below. The model was trained on augmented rice seed images and showed relatively lower accuracy compared to other models. The figure presents the training and validation accuracy and loss over 30 epochs, clearly reflecting the model's learning behavior and areas where underfitting or performance limitations may exist.

EfficientNetB3:

While working with the EfficientNetB3 model, I obtained the results displayed in the figure below. The model was trained using augmented rice seed images and achieved noticeably higher validation accuracy than EfficientNetB0. The training and validation accuracy and loss curves shown in the figure demonstrate steady learning progression over 30 epochs, indicating better feature extraction capability and generalization performance of EfficientNetB3 in rice seed classification.

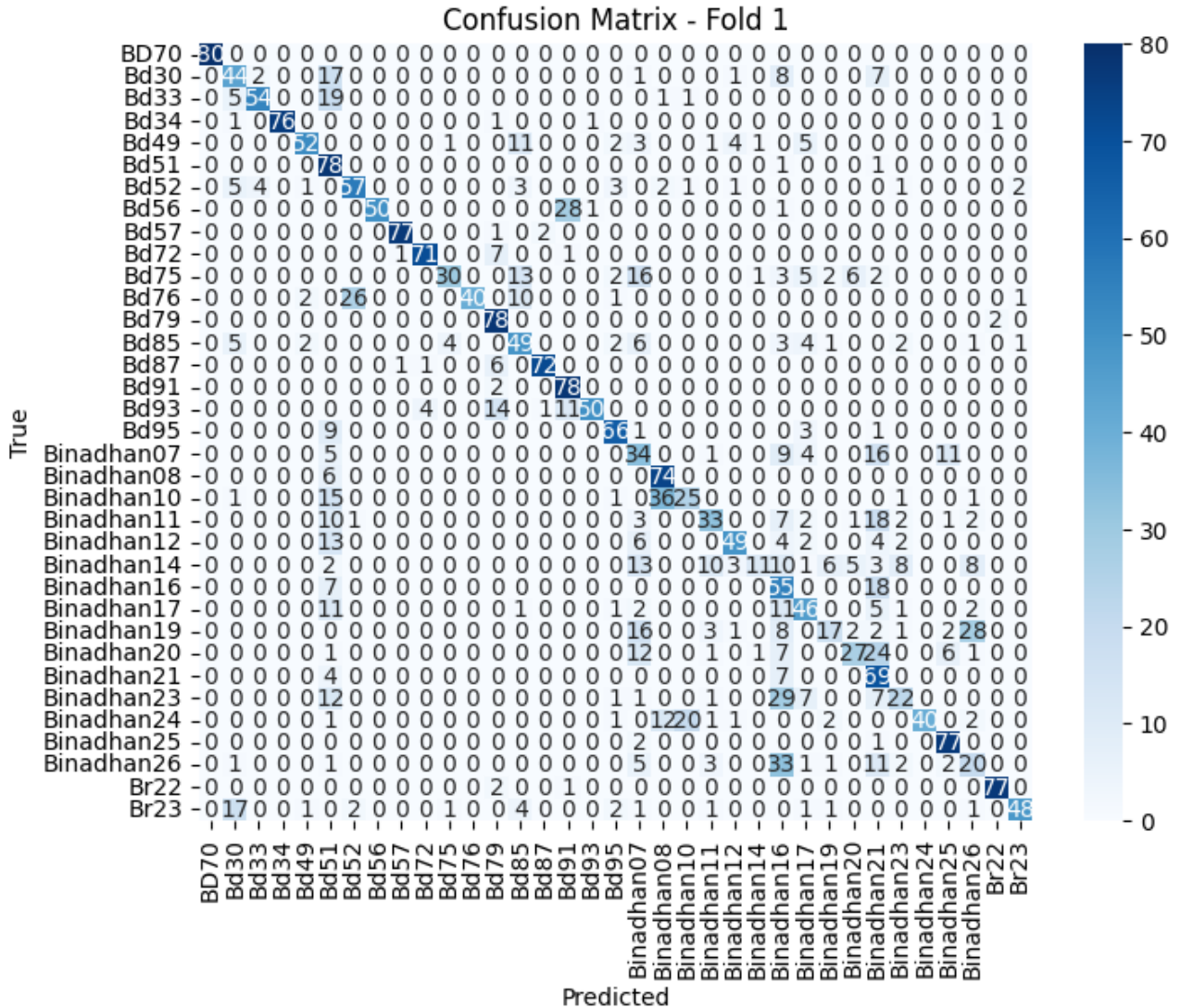


Fig 4: Confusion Matrix

4.3.1 Interpretation of Results

The EfficientNetB3 model achieved an accuracy of 66%, along with balanced precision, recall, and F1-score values, demonstrating its ability to classify various rice seed varieties with moderate reliability. Although the performance was not as high as traditional CNN models like the sequential CNN, EfficientNetB3 benefited from its compound scaling strategy to extract richer visual features. The model, being pre-trained on ImageNet and fine-tuned on the rice seed dataset, leveraged deep feature representations to distinguish among similar-looking seed categories. Grad-CAM visualizations were also used to interpret model predictions. The heatmaps confirmed that EfficientNetB3 focused on biologically relevant seed characteristics—such as texture, edge contours, and shape irregularities—during classification. This interpretability validates that the model was not learning irrelevant noise but meaningful patterns critical for accurate seed variety recognition.

4.3.2 Model Strengths and Weaknesses

Strengths:

- ✓ Utilized transfer learning from ImageNet, enabling better feature extraction even with limited training data.
- ✓ Compound scaling allowed the model to balance depth, width, and resolution efficiently.
- ✓ Performed notably better than EfficientNetB0 and showed steady validation accuracy (~66%) despite dataset complexity.
- ✓ Grad-CAM visualizations confirmed the model's attention to meaningful seed traits like shape and texture.

4.3.3 Limitations of the Study

Although the EfficientNet models and Sequential CNN showed promising results in rice seed classification, this study was conducted on a relatively clean and balanced dataset. However, real-world applications may face several challenges such as:

- **Class imbalance:** Some seed varieties may be overrepresented or underrepresented in field scenarios, affecting generalization.
- **Low-quality or noisy images:** Blurred, occluded, or poorly lit seed images may degrade performance.
- **Lack of multi-modal data:** Only image-based features were used, whereas integrating auxiliary information like seed moisture level, harvest region, or grain weight could enhance accuracy.
- **Controlled environment testing:** The training and evaluation were performed on Kaggle's cloud GPU environment, which does not replicate the variability in lighting, image background, or camera quality in practical field use.
- **Limited exploration:** Due to time and compute constraints, only a limited number of custom and pre-trained models were tested. Broader hyperparameter tuning or ensembling might lead to stronger, more robust models.

4.3.4 Conclusion

This research demonstrates the potential of deep learning techniques in automating rice seed variety classification. Among the tested models, the **Sequential CNN** achieved the highest accuracy of **97.06%**, while **EfficientNetB3** showed improved performance over **EfficientNetB0**, validating the benefits of compound scaling and pre-trained architectures. Despite limited data and training time, the results highlight that even lightweight or custom CNNs can learn valuable discriminative features for rice seed classification. The interpretability provided by **Grad-CAM** confirmed that models focused on biologically meaningful seed attributes, proving their reliability. Overall, deep learning offers a scalable and impactful approach for improving automation in seed quality assessment and classification systems.

4.3.5 Future Works

To enhance the system's reliability and practical usability, future work should consider:

Expanding the dataset with real-world seed images under various lighting conditions, backgrounds, and device cameras to improve generalization.

Developing a mobile application for real-time, on-field seed classification and farmer-friendly diagnosis.

Exploring advanced architectures such as **Vision Transformers (ViT)**, **ConvNeXt**, or hybrid **CNN-Transformer** models for improved performance and explainability.

Integrating external data sources like soil type, region of cultivation, and seasonality could aid in more informed predictions.

Field testing in real agricultural environments to evaluate model robustness under operational constraints.

Applying **automated hyperparameter tuning** or **ensemble methods** to further optimize accuracy and minimize prediction errors.

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